Geometry & Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circle with centre O has radius $r = 8$ cm. Chord AB subtends a central angle $\angle AOB = \frac{3\pi}{4}$ radians. (a) Find the exact length of the minor arc AB. (b) Find the exact area of the sector OAB. (c) Find the exact area of the minor segment enclosed between chord AB and arc AB.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = e^{2x} - 4e^{x} + 3$. Let $u = e^{x}$. Find the minimum value of the function in terms of u.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The first term of a geometric sequence is $u_1 = 18$ and the common ratio is $r = \frac{1}{3}$. (a) Find the exact value of the 5th term u_5. (b) Calculate the sum of the first 6 terms, S_6. (c) State why the sum to infinity S_∞ exists, and find its exact value.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Group A has 20 data values with mean 15. Group B has 18 data values with mean 10. Find the mean of the combined data set.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Points $A(1, 5)$ and $B(-2, 0)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

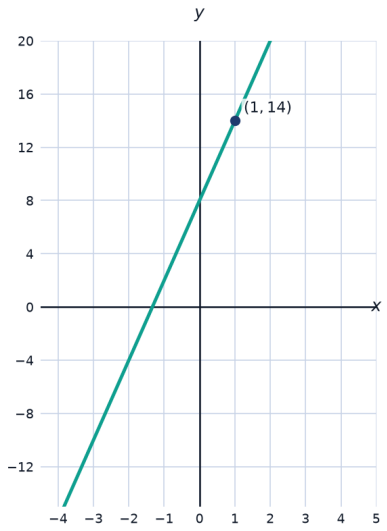
6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The area of a sector is 3 times its arc length for a circle of radius r. Find r.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

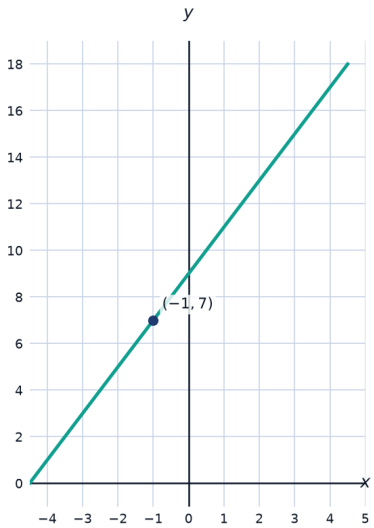
For the model $f(x) = 6x + 8$ in applications of exponential functions, evaluate $f(5)$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(2)$.



9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for the natural exponent consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Practice 3.1: Solve the equation $2\cos 2x = 4\cos x - 3$ for $0 \le x \le 2\pi$, giving exact solutions.

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11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Practice 3.2: Ship A leaves port P on a bearing of 040° traveling at 15 km/h. At the same time, ship B leaves P on a bearing of 130° traveling at 20 km/h. Find the distance between the two ships after 2 hours.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Geometry & Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Practice AI.1: Use the trapezoidal rule with 4 subintervals to estimate $\int_1^3 \sqrt{1 + x^3}\,dx$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Practice 1.1: In an arithmetic sequence, the 4th term is 19 and the sum of the first 10 terms is 290. Find the first term u_1, the common difference d, and the sum of terms from u_{11} to u_{25} inclusive.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Functions

Practice 5.2: A particle moves in a straight line with velocity $v(t) = 3t^2 - 12t + 9$ m/s for $0 \le t \le 4$. (a) Find the times at which the particle is momentarily at rest. (b) Calculate the net displacement of the particle over $[0, 4]$. (c) Calculate the total distance travelled by the particle over $[0, 4]$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Statistics & Probability

Practice 5.1: A curve has equation $y = x^3 - 3x^2 - 9x + 5$. (a) Find the coordinates of the local maximum and local minimum points. (b) Determine the coordinates of the point of inflection and find the equation of the tangent at the point of inflection.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Calculus

Practice AI.2: A company takes out a loan of 120,000 to purchase equipment at an annual interest rate of 6% compounded monthly, repayable by equal monthly payments over 10 years. (a) Find the amount of each monthly payment. (b) Calculate the total interest paid over the life of the loan.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 AI SL — Specific Topics

Practice 1.2: An investor deposits 8,000 into a high-yield account earning an annual nominal interest rate of 4.8% compounded monthly. Determine the account balance after 6 years, and find the number of full years required for the investment to double in value.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number & Algebra

Practice 2.2: The graph of $y = f(x)$ undergoes three consecutive transformations: (1) a horizontal stretch by scale factor 3, (2) a reflection in the y-axis, and (3) a translation by vector $(4;\ -5)$. Write down the equation of the resulting function $g(x)$ in terms of f.

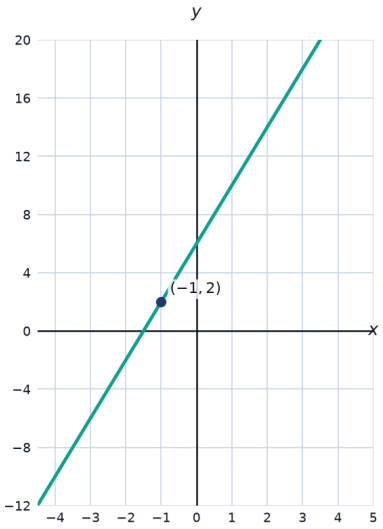
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Functions

An IB AA SL student scores 64/80 on Paper 1 (40%), 72/80 on Paper 2 (40%), and 16/20 on the Internal Assessment (20%). Calculate the student's overall weighted percentage.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 6$ in rational exponents, evaluate $f(2)$.



a. 14

b. 18

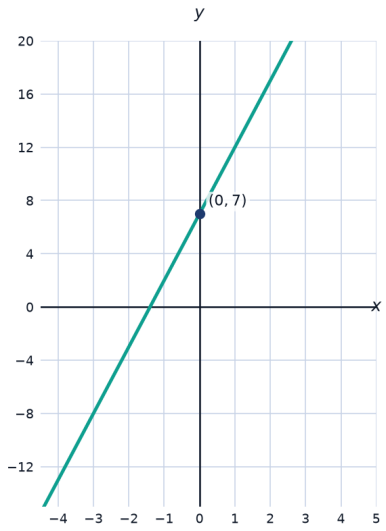
c. 8

d. 18

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 7$ in exponential functions, evaluate $f(2)$.



a. 17

b. 22

c. 10

d. 21

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for rational exponents consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for exponential functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for applications of exponential functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for logarithms consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 3$ in logarithms, evaluate $f(3)$.

a. 12

b. 15

c. 9

d. 16

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for logarithm laws and change of base consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 4x + 4$ in logarithm laws and change of base, evaluate $f(3)$.

a. 16

b. 20

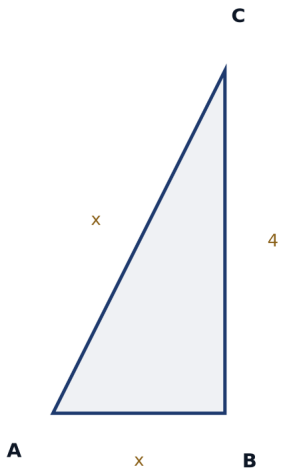
c. 12

d. 20

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In the diagram, triangle ABC is shown with coordinates A(2, 1), B(7, 1), and C(5, 5). What is the length of the side AC?



a. 4

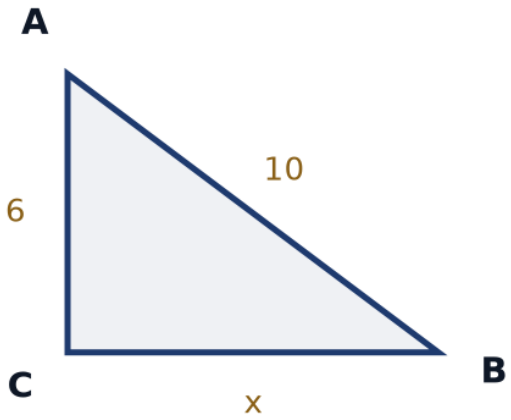
b. 5

c. $2\sqrt{5}$

d. $\sqrt{20}$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the right-angled triangle ABC shown below, the length of side AB is 10 units and the length of side BC is 6 units. The angle ACB is 90 degrees. Find the length of side AC.



a. 10

b. 6

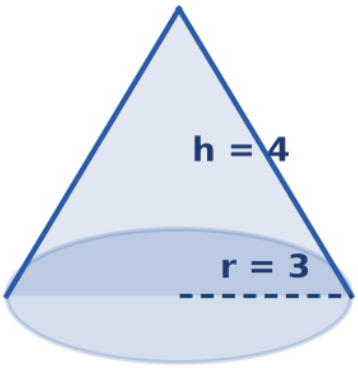
c. 8

d. 16

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Volume of cone with radius 3, height 4:



a. 12π

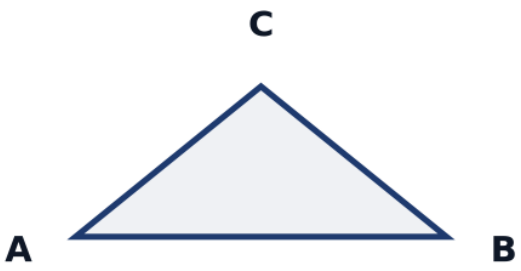
b. 16π

c. 36π

d. 48π

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle ABC, $A = 39^\circ$, $B = 39^\circ$, and $a = 13$. Find b, correct to 1 decimal place.



a. 13.0

b. 15.0

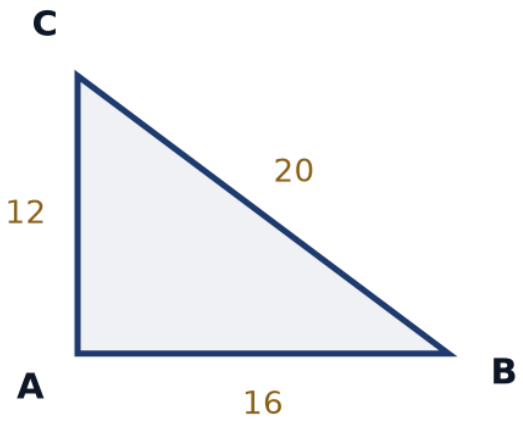
c. 14.0

d. 11.0

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A right triangle has legs 12 and 16 and hypotenuse 20. The angle θ is opposite the side of length 12. Find $\cos(\theta)$.



a. $\frac{4}{3}$

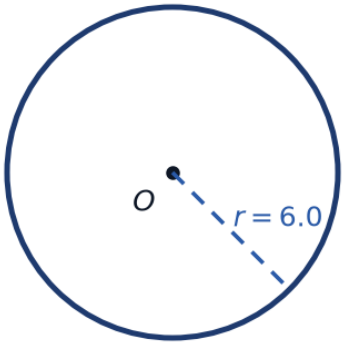
b. $\frac{3}{5}$

c. $\frac{3}{4}$

d. $\frac{4}{5}$

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A sector of a circle with radius 6 cm has area 27 cm^2. Find the angle of the sector, in radians.



a. 1.5

b. 3

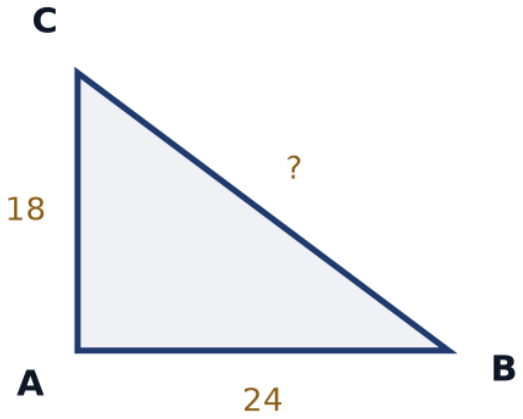
c. 0.75

d. 2

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A right triangle has legs of length 18 and 24. Find the length of the hypotenuse.



a. 32

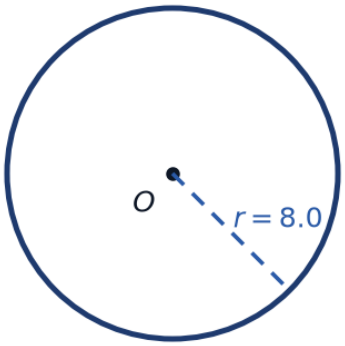
b. 34

c. 33

d. 30

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A circle has radius 8 cm. Find the arc length subtended by an angle of 1.5 radians.



a. 12 cm

b. 6 cm

c. 24 cm

d. 9.6 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 AI SL — Specific Topics

In the diagram, points A, B, and C form a triangle. Point D lies on AC such that BD is the angle bisector of angle ABC. Given that AB = 10 cm, AC = 15 cm, and BC = 12 cm, and that AD = 6 cm, find the length of DC.

a. 8 cm

b. 7.2 cm

c. 9 cm

d. 6 cm

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number & Algebra

In the given right-angled triangle ABC, angle B is 90 degrees. Point D is on BC such that BD = 4 and DC = 9. If AD is the altitude to the hypotenuse AC, find the length of AD.

a. 6

b. 8

c. 12

d. 15

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Functions

In the diagram, triangle ABC has angle ABC equal to 90 degrees. Point D lies on AC such that BD is perpendicular to AC. If AD = 4 cm and DC = 9 cm, find the length of BD.

a. 5 cm

b. 6 cm

c. 7 cm

d. 3 cm

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Statistics & Probability

In the given triangle ABC, AB = 10, AC = 15, and BD = 6. D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

a. 9

b. 10

c. 15

d. 22.5

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Calculus

In the right-angled triangle ABC, shown below, AD is the altitude to the hypotenuse BC. If BD = 4 and DC = 9, find the length of the altitude AD.

a. 5

b. 6

c. 7

d. 8

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 AI SL — Specific Topics

In the triangle ABC, AB = 10, AC = 15, BD = 6, and D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

a. 7

b. 9

c. 10

d. 12

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Number & Algebra

A sector of a circle has radius 9 cm and angle 45°. Find the exact area of the sector.

a. $\frac{81\pi}{8}$

b. $\frac{49\pi}{8}$

c. 8π

d. $\frac{9\pi}{2}$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Functions

In triangle ABC, $a = 5$, $b = 10$, and $C = 120^\circ$. Find the exact length of c.

a. $5\sqrt{7}$

b. $\sqrt{179}$

c. 15

d. $3\sqrt{19}$

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